

CL papers AI agent

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Graph build (03), neo4j

[update previous tasks chat](#), [Neo4j chat](#)

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Evaluation (04), fastapi demo (05)

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Ingestion pipeline (01)

[ingestion chat](#)

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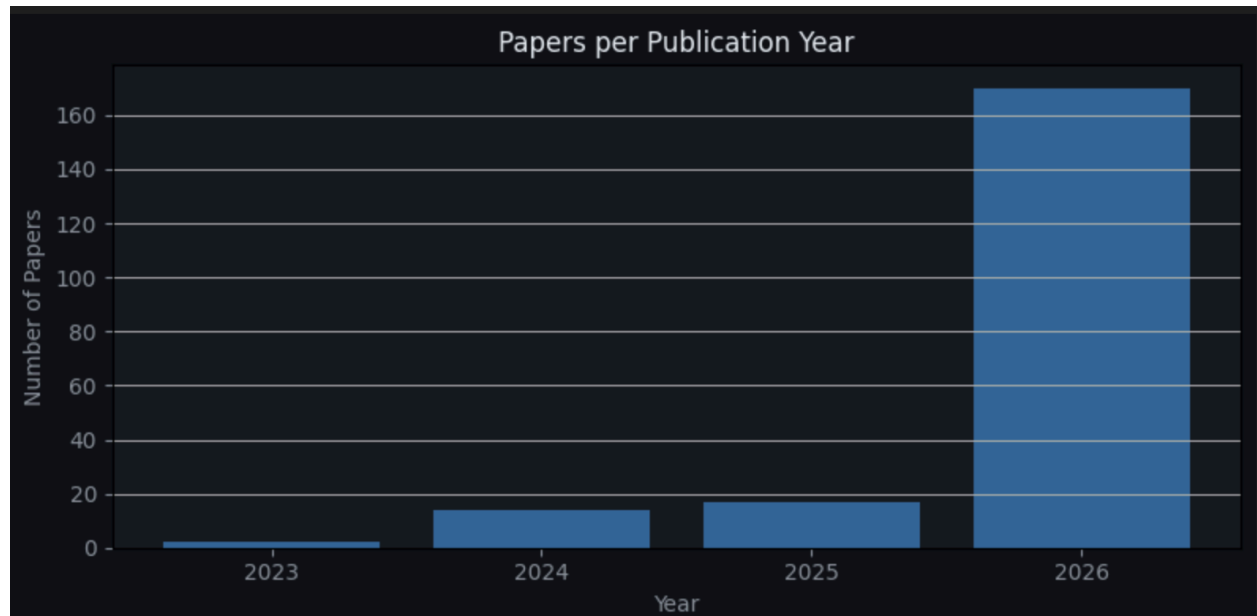
Docker, semantic retrieval (02), MongoDB

[semantic chat](#)

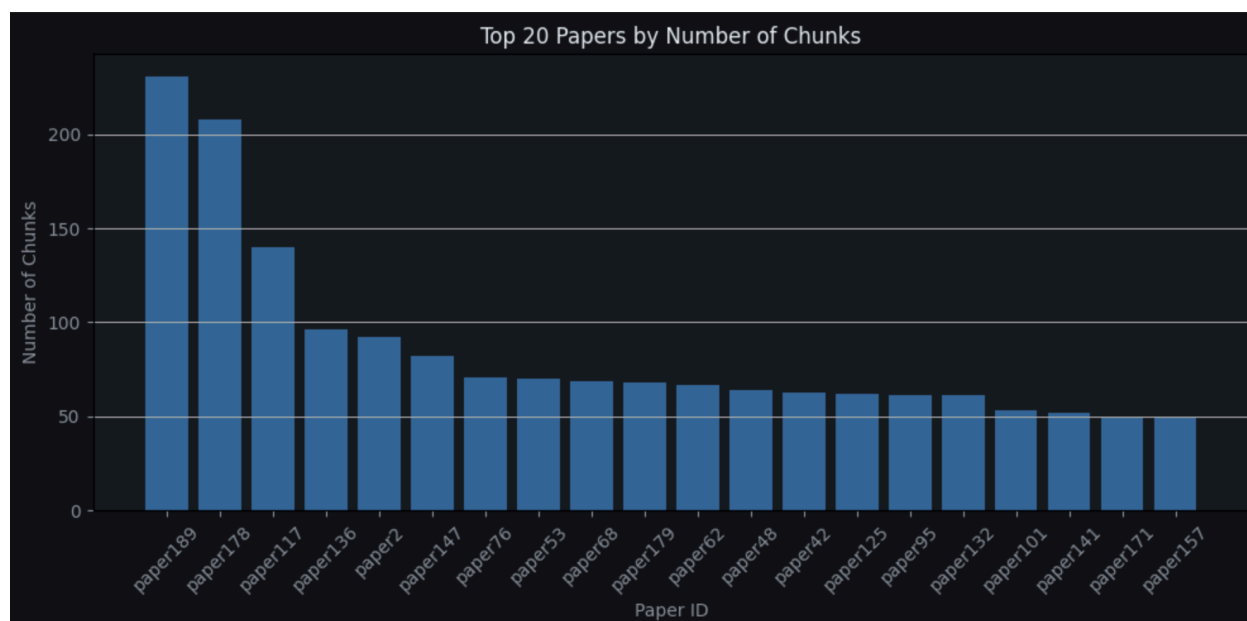
Youtube Demo Link: <https://youtu.be/T60rS1St0m4>

Ingestion Pipeline (Notebook 01)

The ingestion pipeline is the first preparatory stage of CLPapersAIAgent. It consists of validating and organizing the input PDF files, extracting the bibliographical metadata using PyMuPDF, segmenting the text into overlapping 400-word segments, and saving it all to MongoDB (clpapers.chunks).



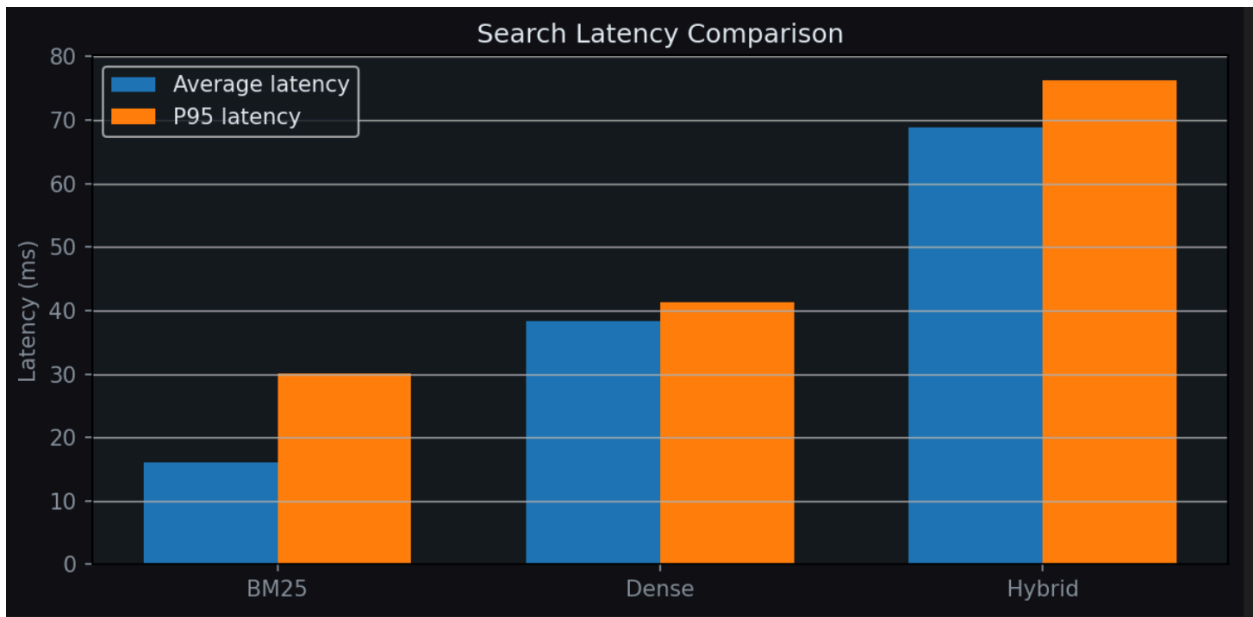
Distribution of papers across the years of publication is shown in the chart above. There is a huge skewness in the data set favoring the year 2026, having 170 papers, followed by 17 papers from 2025, 14 papers from 2024, and only 2 from 2023.



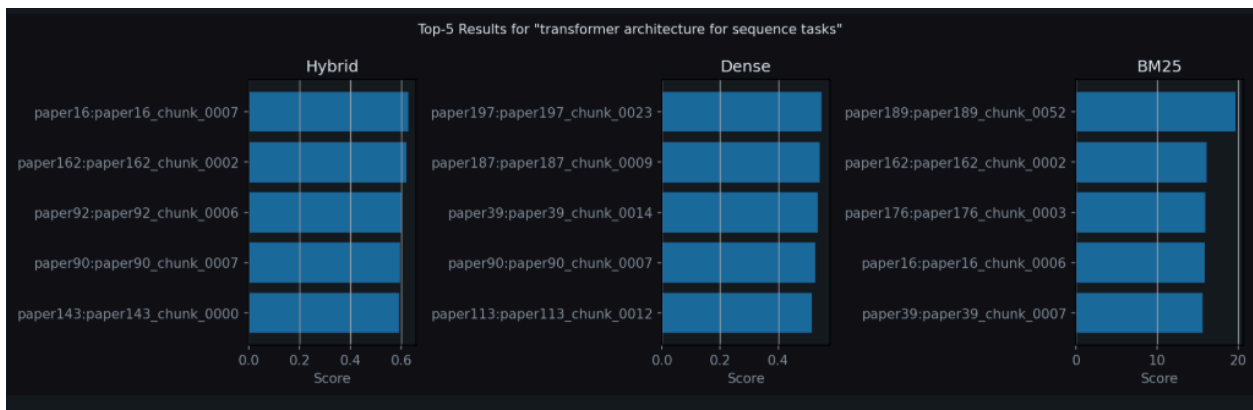
Bar Chart showing the 20 top-ranked papers according to the number of chunks that were created after ingesting the paper. Paper189 and paper178 stand out by having more than 200 chunks created from them, whereas the other papers decline progressively, most being in the 50-100 range. This is because there was a variation in the length of the papers in the dataset.

Semantic Retrieval (Notebook 02)

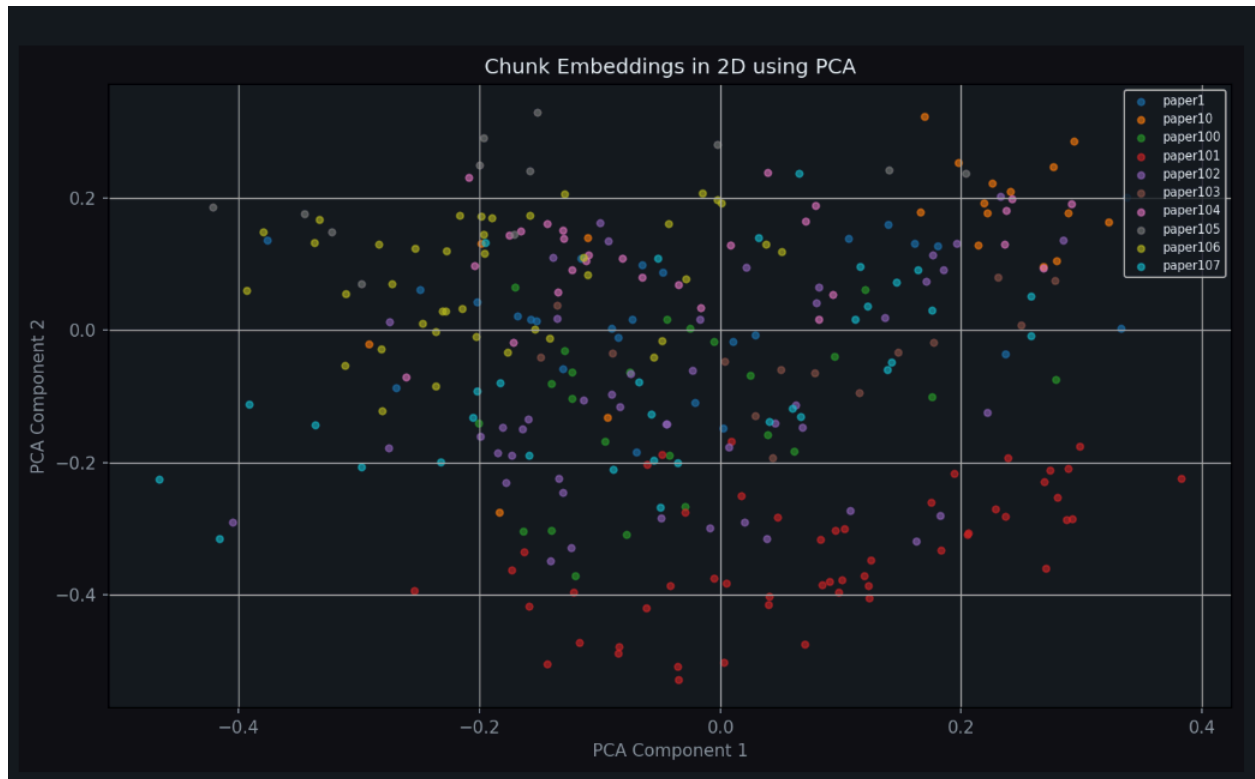
Bar Chart



This chart compares the average latency and P95 latency of BM25, Dense, and Hybrid search. Average latency shows the typical response time, while P95 latency shows the slower cases. BM25 is usually fastest because it uses keyword-based matching, while Dense and Hybrid search take longer because they rely on vector similarity and additional retrieval processing.

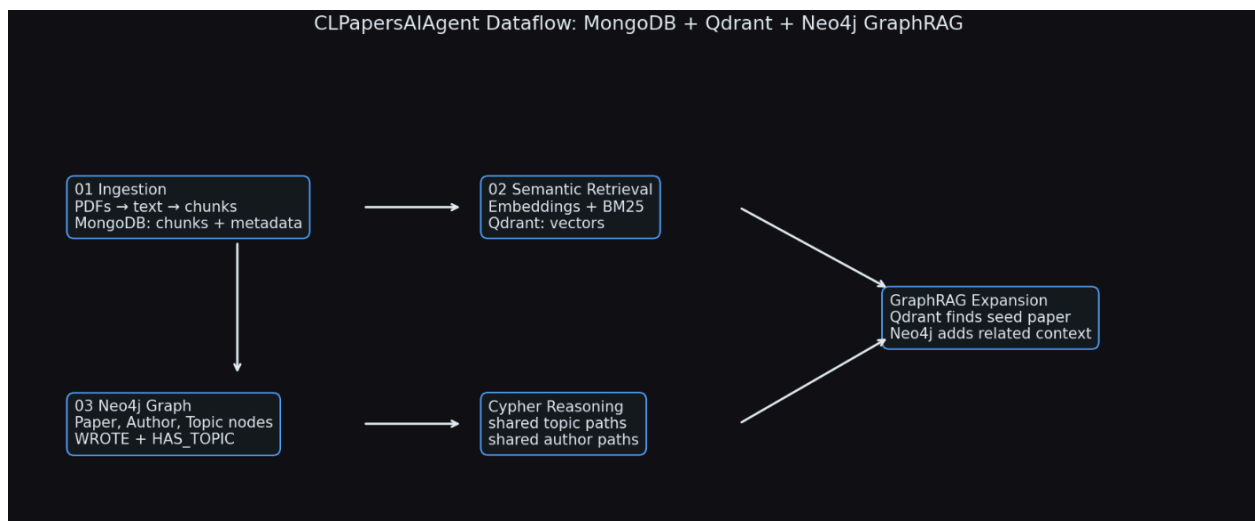


This chart shows an example of the top five retrieved chunks returned by BM25, Dense, and Hybrid for the same query. It helps compare how each method ranks its results and whether the retrieved chunks appear relevant to the query. This is useful for qualitative evaluation, not just numerical metric comparison.



This visualization shows high-dimensional chunk embeddings reduced into 2D using PCA. Each point represents a text chunk from a research paper, and colors represent different papers. Nearby points suggest that chunks have similar semantic meaning. This helps explain how Dense retrieval can find related content based on meaning rather than exact keyword matching.

Graph Build (Notebook 03)

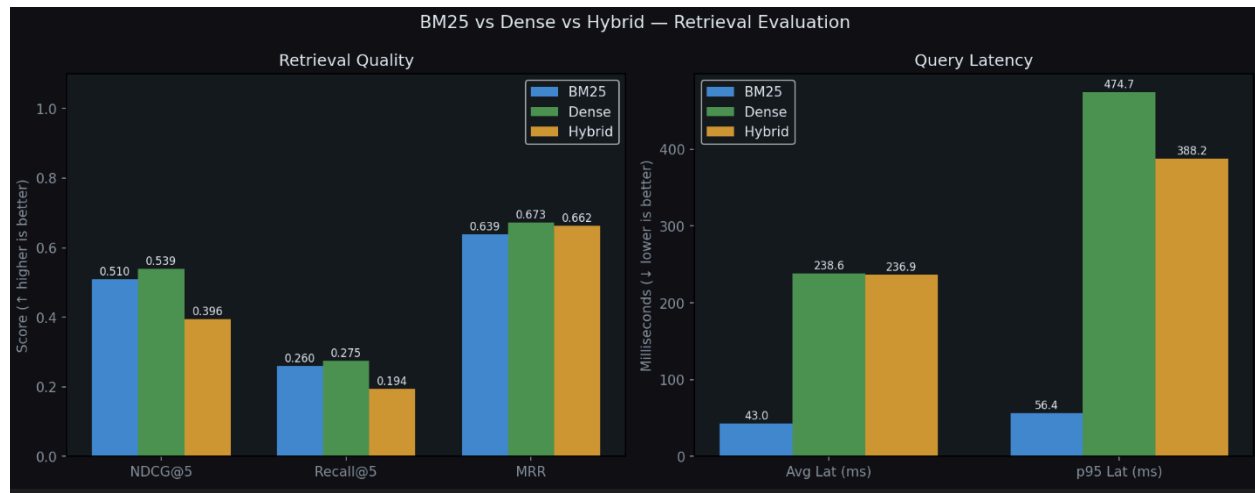


This diagram shows how the full retrieval system is connected. The ingestion stage extracts text, chunks, and metadata from PDFs. MongoDB stores the raw chunks and metadata, Qdrant stores

vector embeddings for semantic search, and Neo4j stores the knowledge graph. The graph connects Paper, Author, and Topic nodes through relationships such as WROTE and HAS_TOPIC, allowing GraphRAG to add related context beyond the initially retrieved results.

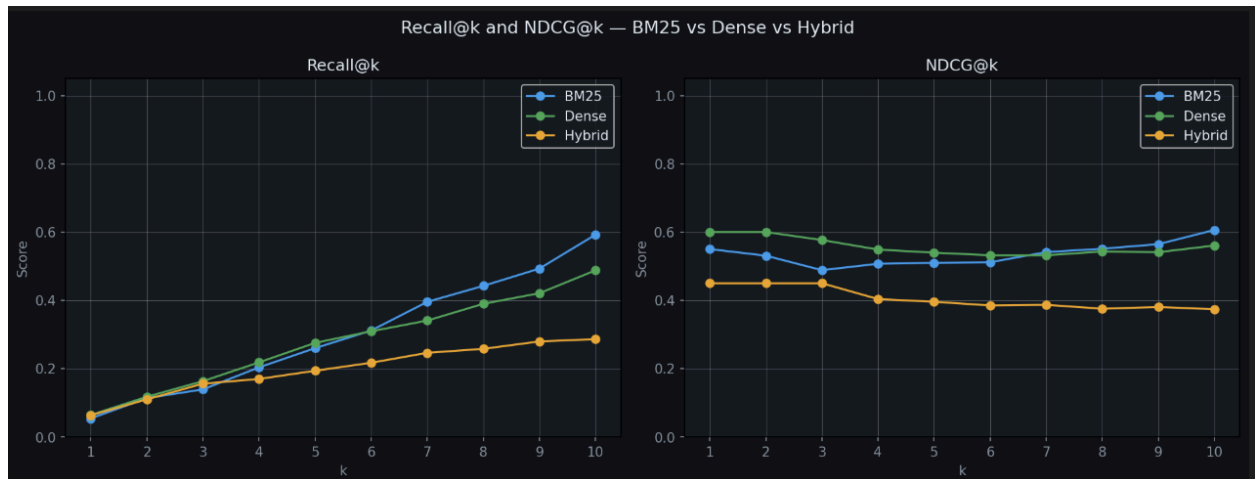
Evaluation (Notebook 04)

Bar Chart



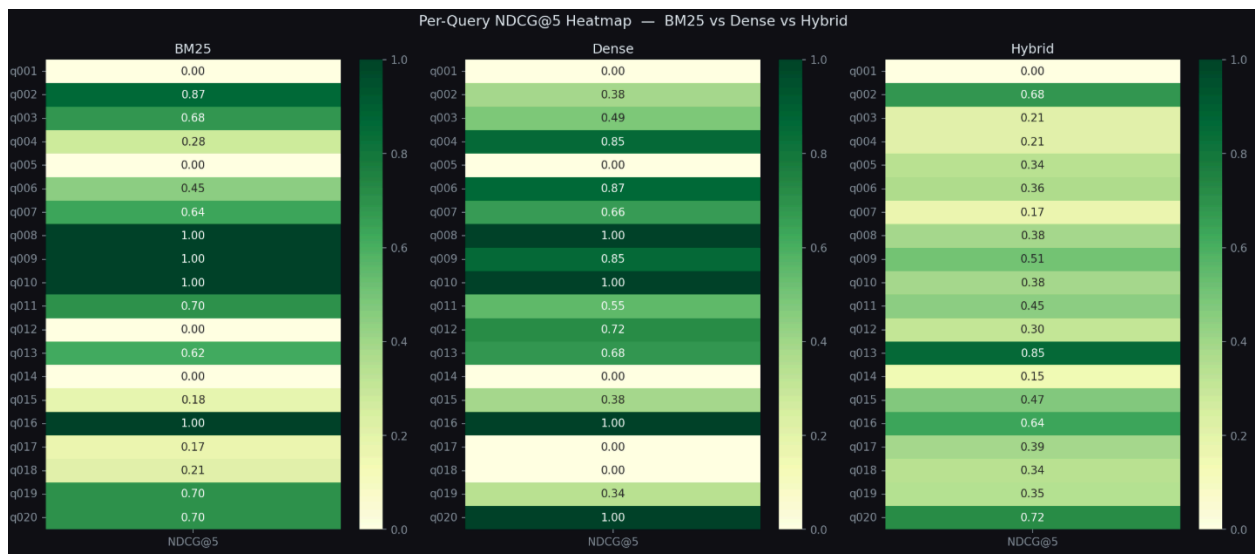
This chart compares BM25, Dense, and Hybrid retrieval using Recall@5, NDCG@5, MRR, average latency, and P95 latency. The quality chart shows how well each method retrieves and ranks relevant papers, while the latency chart shows how fast each method responds. Dense performs strongly in ranking quality, BM25 is the fastest, and Hybrid provides a combined retrieval approach using both lexical and semantic signals.

Line Chart



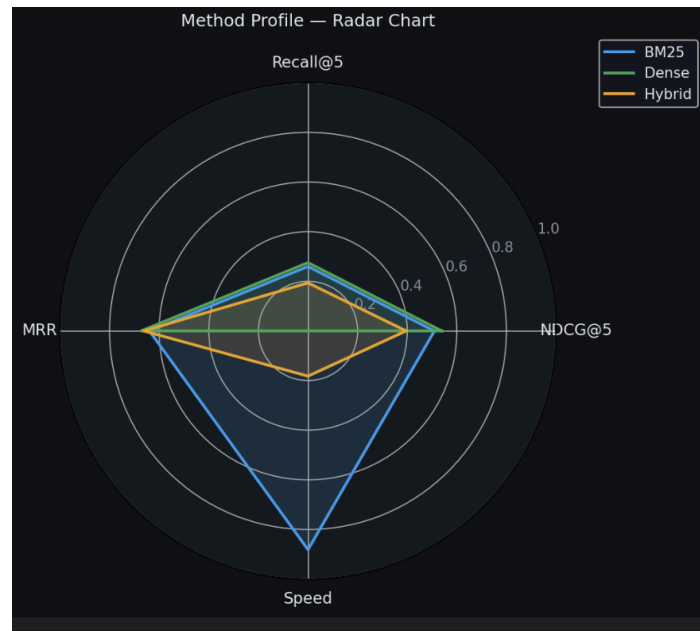
These curves show how retrieval performance changes as k increases from 1 to 10. Recall@ k measures how many relevant papers are found as more results are returned, while NDCG@ k measures whether the most relevant papers appear near the top. A method with higher curves is better at finding useful papers earlier in the ranked list.

Heatmap



This heatmap shows the NDCG@5 score for each query across BM25, Dense, and Hybrid retrieval. Each row represents one query, and darker green cells indicate stronger ranking performance. The chart helps identify which queries each method handles well and which query types are more challenging for a specific retrieval approach.

Radar Chart



This radar chart summarizes each retrieval method across multiple factors at the same time: Recall@5, NDCG@5, MRR, and speed. A larger shape means stronger overall performance. BM25 has an advantage in speed, Dense performs well on semantic quality metrics, and Hybrid shows how combining methods can balance retrieval effectiveness with search performance.

Future Improvements & Context

The extraction of metadata may not work for scans since OCR and other APIs, such as CrossRef, will be a more robust solution to this problem. Chunking into fixed-sized segments is independent of the document's structure, and therefore, splitting with awareness of the sections would result in improved results.